

ATPL EASA STUDY GUIDE | 081 - PRINCIPLES OF FLIGHT

Chapter 10: Static & Dynamic Stability Fundamentals

Syllabus Source: CAE Oxford (pp. 345-384) | Pass Target: >= 90% AviationExam

1. STATIC STABILITY: INITIAL TENDENCY UPON DISTURBANCE

Stability defines an aircraft's response to an aerodynamic disturbance from trimmed equilibrium:

- Positive Static Stability: The INITIAL tendency to return toward original trimmed equilibrium following disturbance (creates an opposing restoring moment: $dC_M / d\alpha < 0$).
- Neutral Static Stability: The initial tendency to remain in the displaced position with zero restoring or diverging force ($dC_M / d\alpha = 0$).
- Negative Static Stability (Static Instability): The initial tendency to diverge further away from trimmed equilibrium ($dC_M / d\alpha > 0$).

2. DYNAMIC STABILITY: BEHAVIOR OVER TIME

Dynamic stability describes the time history of the motion following an initial statically stable response. Dynamic stability CAN ONLY EXIST if the aircraft possesses positive static stability first!

Dynamic Stability Condition	Time-History Motion Characteristics	Physical Restoring & Damping Forces
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Positive Dynamic Stability	Oscillations decrease in amplitude over time (damped motion) until steady equilibrium is restored.	Aerodynamic damping (e.g. tailplane pitch damping) dissipates kinetic energy into air.
Neutral Dynamic Stability	Oscillations continue indefinitely with CONSTANT amplitude (undamped sine wave).	Zero net damping force.
Negative Dynamic Stability	Oscillations amplify progressively over time with growing amplitude (divergent oscillation)!	Energy is continuously pumped into the oscillation, leading to structural divergence.

3. LONGITUDINAL DYNAMIC MODES: SPPO VS PHUGOID

Disturbing an aircraft longitudinally excites two distinct coupled pitch-speed oscillations:

- Short Period Pitch Oscillation (SPPO): High frequency (period typically 1 to 2 seconds), heavily damped. Consists of rapid pitch angle and Angle of Attack oscillations at virtually constant airspeed! If under-damped or if pilot attempts manual corrections out-of-phase, can trigger catastrophic Pilot Induced Oscillations (PIO)!
- Phugoid Motion (Long Period Mode): Low frequency (period typically 20 to 60+ seconds), weakly damped. Consists of a slow exchange between kinetic energy (airspeed) and potential energy (altitude) at VIRTUALLY CONSTANT ANGLE OF ATTACK! Aircraft climbs and slows down, then dives and speeds up. Easily controlled by pilot or autopilot altitude hold.

[!] EXAM TRAP (AVIATIONEXAM) - Short Period (SPPO) vs Phugoid Distinction Trap

- Short Period Pitch Oscillation: Variable AOA, CONSTANT airspeed! (High frequency, dangerous for PIO). - Phugoid Oscillation: CONSTANT AOA, variable airspeed and altitude! (Low frequency, easily damped by pilot). - Strict Rule: An aircraft CANNOT be dynamically stable unless it is statically stable first!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Static vs Dynamic Stability**

Exam Problem: A gust displaces the nose up by 5 degrees. The aircraft pitches down to recover, but overshoots and begins oscillations of 6, 8, and 10 degrees. Classify its stability. Step 1 (Initial Action): It initiated a return to equilibrium -> POSITIVE STATIC STABILITY. Step 2 (Time History): The oscillations increased in amplitude (6, 8, 10) -> DIVERGENT -> NEGATIVE DYNAMIC STABILITY. Conclusion: The aircraft is statically stable but dynamically unstable.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the defining characteristic of an aircraft exhibiting POSITIVE STATIC STABILITY?**

- [A]: It oscillates indefinitely at constant amplitude
- [B]: Its INITIAL tendency following an aerodynamic disturbance is to develop forces that restore it toward original trimmed equilibrium
- [C]: It requires no pilot controls
- [D]: It flies at supersonic speed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Static stability describes purely the immediate initial tendency upon release from a disturbed displacement. Positive static stability means restoring moments oppose the disturbance.

Q2: In longitudinal dynamic flight modes, how do angle of attack and airspeed behave during a classic PHUGOID mode?

- [A]: Angle of attack oscillates rapidly while airspeed remains constant
- [B]: Angle of attack remains virtually CONSTANT while airspeed and altitude oscillate slowly
- [C]: Both parameters remain completely frozen
- [D]: Pitch damping is zero

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The phugoid mode is a long-period trade between kinetic and potential energy. Because pitch attitude tracks the flight path, the angle of attack remains essentially constant while speed and altitude cycle.

Q3: Can an aircraft possess positive dynamic stability if it has negative static stability?

- [A]: Yes, always
- [B]: NO, dynamic stability requires positive static stability as a prerequisite
- [C]: Only at high Mach numbers
- [D]: Only with flaps down

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. If an aircraft is statically unstable, it immediately diverges away from equilibrium, preventing any oscillatory motion from developing. Therefore, positive static stability is mandatory for dynamic stability.